

Harsh Raj



Education

MS in Computer Science, *Northeastern University*, Boston, US.

2025–Present

B.Tech in Engineering Physics, *Delhi Technological University*, New Delhi, India.

2019–2023

Grade: 8.35/10.0

Skills Summary

Languages: Python, Java, C++, C, SQL, Unix scripting

Frameworks & Tools: Kubernetes, Docker, GIT, Matlab, Tensorflow, Pytorch, FastAPI

Experience

Co-Founder, Ontocord AI (Open Source), California, US.

Oct 2024 – Present

- Developed a pipeline for multimodal data synthesis from scratch using a novel approach based on ontological verb-object perturbation.
- Optimized the pipeline to support multi-node multiprocessing for parallel execution, achieving a 40x speedup. This optimization is designed to run efficiently on the [Leonardo](#) supercomputer.

AI Researcher (Contract) (Remote), *AI Risk and Vulnerability Alliance (ARVA)*, Washington, US.

Oct 2024 – Present

- Developing “bias-inspect,” an open-source toolkit for benchmarking Large Language Models (LLMs) for various biases, including gender, racial, political, and age-related biases.
- Building a flexible API to facilitate the easy integration of new benchmarks.

Applied Scientist (full-time) (remote), *Vijil AI*, California, US.

Feb 2024 - Oct 2024

- Developed [vijil-fuzzer](#), an LLM red-teaming framework that mutates seed prompts for aggressive behavior and jailbreak scenarios.
- Integrated over 30 open-source benchmarks into the [vijil-evaluation](#) service.
- Contributed to [vijil-dome](#) and [guardrails](#), integrating guardrails into tools like Langchain with minimal latency impact.

ML Engineer (full-time) (remote), *Yield Protocol*, Chicago, US.

Sept 2022 - Jan 2024

- Led data synthesis to improve LLMs as agents and built [Synchaeu](#), a complete system for agent data generation.
- Developed [Mandrill](#), a framework for fine-tuning LLMs and validation on agent benchmarks. Supervised 10 research interns from the [Disruption Lab](#) at UIUC as part of the project.
- Co-created [Cacti](#), a web3 transaction chatbot, in collaboration with a former VP of Quora.
- Built [AutoEval](#), a framework for automating the evaluation of the Cacti bot.

Applied Researcher (intern) (remote), *Thoucentric*, Bangalore, India.

May 2022 - Aug 2022

- Developed a comprehensive NL2SQL conversion system tailored for data analysis purposes. Incorporated automatic visualization feature with tools like [DeepEye](#).
- Improved the [SADGA-GaP](#) framework by introducing pre-processing and post-processing steps, resulting in a 10% increase in accuracy. Additionally, implemented a value-copying mechanism in the model, resolving issues such as missing table names and row values in generated SQL queries.
- Authored a [white paper](#)

Data Scientist (intern) (on-site), *Attryb Tech*, Bangalore, India.

June 2021 - Dec 2021

- Developed the complete pipeline for [Content Studio](#), a tool for generating and analyzing content for blog composition, product marketing, and various other creative writing tasks.
- Finetuned Pegasus and GPT-2 for content summarization and title generation.
- Data mined [SemRush](#) and [Common Crawl](#) to create the content database.

Publications

[Improving Consistency in Large Language Models through Chain of Guidance](#) (TMLR'2024): Harsh Raj, Vipul Gupta, Domenic Rosati, Subhabrata Majumdar

[Defences against Reverse Preference Attacks](#) (AI Security Workshop, AAAI'25): Domenic Rosati, Giles Edkins, Harsh Raj, David Atanasov, Kai Williams, Subhabrata Majumdar, Janarthanan Rajendran, Frank Rudzicz, Hassan Sajjad

[On Transfer of Adversarial Robustness from Pretraining to Downstream Tasks](#) (NeurIPS'23): Laura Fee Nern, Harsh Raj, Maurice Georgi, Yash Sharma

[Measuring Reliability of Large Language Models through Semantic Consistency](#) (Best Paper Award, ML Safety Workshop, NeurIPS'22): Harsh Raj, Domenic Rosati, Subhabrata Majumdar

[Evaluating the Robustness of Biomedical Concept Normalization](#) (Transfer Learning Workshop, NeurIPS'22): Sinchani Chakraborty, Harsh Raj, Srishti Gureja, Tanmay Jain, Atif Hassan, Sayantan Basu

[Decoding Percepts in Vision Language Navigation: Is it about better features or more data?](#) (under review, ACM TIST): Harsh Raj, Ashutosh Pandey, Shaurya Kumar, Kavinder Singh, Nihal Kumar, Anil Singh Parihar

[GANDALF: Gated Adaptive Network for Deep Automated Learning of Features](#) (under review, TMLR): Manu Joseph, Harsh Raj

[AskYourDB: An end-to-end system for querying and visualizing relational databases using natural language](#): Manu Joseph, Harsh Raj, Abhinav Yadav, Aaryamann Sharma

[Extract It! Product Category Extraction by Transfer Learning](#) (CICIT'22): Harsh Raj, Aakansha Gupta, Rahul Katarya

Projects

[Interpreting the Effects of Jailbreaking](#) (Mechanistic Interpretability):

- Understanding the effects of Jailbreaking in LLMs through mechanistic interpretability.

[Synchae](#) (Data Synthesis):

- A framework for generating conversational data between LLM agents and environments. The environments are taken from [AgentBench](#).

[Repo-Level Prompt Engineering-Solidity](#) (Prompt Engineering):

- An implementation of the paper [Repo-Level Prompt Generation](#) to support SOLIDITY code, a widely used programming language in the Blockchain industry.

[Antibody-Antigen Binding Classifier](#) (Computational Biology):

- A graph neural network-based architecture to predict the binding affinity between antibodies and antigens, utilizing the [SAbDab](#) dataset for affinity prediction.

Honors and Awards

Vision and Language Navigation: Secured 3rd place in the SPL metric on the R2R benchmark, a widely recognized Vision and Language Navigation measure. [Link to standings](#). ID: *MLR.Lab.DTU*.

ML Safety Challenge: Won NeurIPS'22 [Best Paper Award](#) with a cash prize of \$5000.

Codeforces ML Round: Achieved **Global Rank 7** and ranked **1st** in the country in the Raif ML Round 1, organized by Raiffeisen Bank International AG. [Link to standings](#). ID: *harsh777111raj*.